

$$K_0 = \int_V \rho \vec{u} \cdot \vec{u}^* dV. \quad (1)$$

$$K_\rho = \int_V \left\{ \delta \rho \left[8\pi G \rho u_r u_r^* - \sigma^2 \vec{u} \cdot \vec{u}^* + g\Lambda + 4\pi G \int_r^R \rho \Lambda dq - (\vec{u} \cdot \nabla \phi_1^* + \nabla \phi_1 \cdot \vec{u}^*) \right] \right\} dV, \quad (2)$$

$$\Lambda = (\vec{u} \cdot \nabla) u_r^* - u_r (\nabla \cdot \vec{u}^*) - \frac{2}{r} u_r u_r^*. \quad (3)$$

$$K_\lambda = \int_V \delta \lambda (\nabla \cdot \vec{u}) (\nabla \cdot \vec{u}^*) dV, \quad (4)$$

$$K_\mu = \int_V \delta \mu \{ \nabla \cdot (2\vec{\epsilon} \cdot \vec{u}^*) - [\nabla \cdot (2\vec{\epsilon})] \cdot \vec{u}^* \} dV. \quad (5)$$

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$$\vec{u} = \left(U_n^k P_n^k \hat{e}_r + V_n^k \frac{dP_n^k}{d\theta} \hat{e}_\theta + V_n^k \frac{ik}{\sin \theta} P_n^k \hat{e}_\lambda \right) e^{ik\lambda}, \quad (6)$$

$$\vec{u}^* = \left(U_n^k P_n^k \hat{e}_r + V_n^k \frac{dP_n^k}{d\theta} \hat{e}_\theta - V_n^k \frac{ik}{\sin \theta} P_n^k \hat{e}_\lambda \right) e^{-ik\lambda}, \quad (7)$$

$$\vec{u} \cdot \vec{u}^* = (U_n^k)^2 (P_n^k)^2 + (V_n^k)^2 \left(\frac{dP_n^k}{d\theta} \right)^2 + (V_n^k)^2 \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \quad (8)$$

$$\int_V \rho \vec{u} \cdot \vec{u}^* dV = 2\pi \int_0^R \int_0^\pi \rho \left[(U_n^k)^2 (P_n^k)^2 + (V_n^k)^2 \left(\frac{dP_n^k}{d\theta} \right)^2 + (V_n^k)^2 \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \right] \sin \theta d\theta dr \quad (9)$$

$$\int_V \rho \vec{u} \cdot \vec{u}^* dV = \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \rho r^2 [(U_n^k)^2 + n(n+1)(V_n^k)^2] dr, \quad (10)$$

$$K_0 = \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \rho r^2 [(U_n^k)^2 + n(n+1)(V_n^k)^2] dr, \quad (11)$$

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$$\int_V \delta \rho \rho u_r u_r^* dV = \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta \rho \rho r^2 (U_n^k)^2 dr, \quad (12)$$

$$\int_V \delta \rho \vec{u} \cdot \vec{u}^* dV = \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta \rho r^2 [(U_n^k)^2 + n(n+1)(V_n^k)^2] dr, \quad (13)$$

$$\begin{aligned} \Lambda = & U_n^k \frac{dU_n^k}{dr} (P_n^k)^2 + \frac{U_n^k V_n^k}{r} \left(\frac{dP_n^k}{d\theta} \right)^2 + \frac{U_n^k V_n^k}{r} \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \\ & - \left[U_n^k \frac{dU_n^k}{dr} + \frac{2(U_n^k)^2}{r} - n(n+1) \frac{U_n^k V_n^k}{r} \right] (P_n^k)^2 - \frac{2(U_n^k)^2}{r} (P_n^k)^2 \end{aligned} \quad (14)$$

$$\Lambda = -\frac{4(U_n^k)^2}{r} (P_n^k)^2 + n(n+1) \frac{U_n^k V_n^k}{r} (P_n^k)^2 + \frac{U_n^k V_n^k}{r} \left(\frac{dP_n^k}{d\theta} \right)^2 + \frac{U_n^k V_n^k}{r} \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \quad (15)$$

$$\int_V \delta \rho \left[g\Lambda + 4\pi G \int_r^R \rho \Lambda dq \right] dV = \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta \rho r^2 \left\{ -2g \left[\frac{2(U_n^k)^2}{r} - n(n+1) \frac{U_n^k V_n^k}{r} \right] \right.$$

$$-8\pi G \int_r^R \rho \left[\frac{2(U_n^k)^2}{q} - n(n+1) \frac{U_n^k V_n^k}{q} \right] dq \Big\} dr \quad (16)$$

$$\vec{u} \cdot \nabla \phi_1^* + \nabla \phi_1 \cdot \vec{u}^* = 2 \left[U_n^k \frac{d\Phi_n^k}{dr} (P_n^k)^2 + \frac{V_n^k \Phi_n^k}{r} \left(\frac{dP_n^k}{d\theta} \right)^2 + \frac{V_n^k \Phi_n^k}{r} \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \right] \quad (17)$$

$$\int_V \delta \rho [\vec{u} \cdot \nabla \phi_1^* + \nabla \phi_1 \cdot \vec{u}^*] dV = \frac{8\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta \rho r^2 \left[U_n^k \frac{d\Phi_n^k}{dr} + n(n+1) \frac{V_n^k \Phi_n^k}{r} \right] dr \quad (18)$$

$$\begin{aligned} K_\rho &= \int_V \left\{ \delta \rho \left[8\pi G \rho u_r u_r^* - \sigma^2 \vec{u} \cdot \vec{u}^* + g\Lambda + 4\pi G \int_r^R \rho \Lambda dq - (\vec{u} \cdot \nabla \phi_1^* + \nabla \phi_1 \cdot \vec{u}^*) \right] \right\} dV \\ &= 8\pi G \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta \rho r^2 (U_n^k)^2 dr \\ &\quad - \sigma^2 \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta \rho r^2 [(U_n^k)^2 + n(n+1)(V_n^k)^2] dr \\ &\quad + \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta \rho r^2 \left\{ -2g \left[\frac{2(U_n^k)^2}{r} - n(n+1) \frac{U_n^k V_n^k}{r} \right] \right. \\ &\quad \left. - 8\pi G \int_r^R \rho \left[\frac{2(U_n^k)^2}{q} - n(n+1) \frac{U_n^k V_n^k}{q} \right] dq \right\} dr \\ &\quad - \frac{8\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta \rho r^2 \left[U_n^k \frac{d\Phi_n^k}{dr} + n(n+1) \frac{V_n^k \Phi_n^k}{r} \right] dr \\ &= \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta \rho r^2 \left\{ 8\pi G \rho (U_n^k)^2 - \sigma^2 [(U_n^k)^2 + n(n+1)(V_n^k)^2] \right. \\ &\quad \left. - 2g \left[\frac{2(U_n^k)^2}{r} - n(n+1) \frac{U_n^k V_n^k}{r} \right] - 8\pi G \int_r^R \rho \left[\frac{2(U_n^k)^2}{q} - n(n+1) \frac{U_n^k V_n^k}{q} \right] dq \right. \\ &\quad \left. - 2 \left[U_n^k \frac{d\Phi_n^k}{dr} + n(n+1) \frac{V_n^k \Phi_n^k}{r} \right] \right\} dr \quad (19) \end{aligned}$$

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$$K_\lambda = \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta \lambda r^2 \left[\frac{dU_n^k}{dr} + \frac{2U_n^k}{r} - n(n+1) \frac{V_n^k}{r} \right]^2 dr \quad (20)$$

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$$\begin{aligned} 2\vec{e} \cdot \vec{u}^* &= [\nabla \vec{u} + (\nabla \vec{u})^T] \cdot \vec{u}^* \\ &= \{ 2(\nabla \vec{u})_{rr} u_r^* + [(\nabla \vec{u})_{r\theta} + (\nabla \vec{u})_{\theta r}] u_\theta^* + [(\nabla \vec{u})_{r\lambda} + (\nabla \vec{u})_{\lambda r}] u_\lambda^* \} \hat{e}_r \\ &\quad + \{ [(\nabla \vec{u})_{\theta r} + (\nabla \vec{u})_{r\theta}] u_r^* + 2(\nabla \vec{u})_{\theta\theta} u_\theta^* + [(\nabla \vec{u})_{\theta\lambda} + (\nabla \vec{u})_{\lambda\theta}] u_\lambda^* \} \hat{e}_\theta \\ &\quad + \{ [(\nabla \vec{u})_{\lambda r} + (\nabla \vec{u})_{r\lambda}] u_r^* + [(\nabla \vec{u})_{\lambda\theta} + (\nabla \vec{u})_{\theta\lambda}] u_\theta^* + 2(\nabla \vec{u})_{\lambda\lambda} u_\lambda^* \} \hat{e}_\lambda. \quad (21) \end{aligned}$$

$$\begin{aligned} \{ 2\vec{e} \cdot \vec{u}^* \}_n^k &= \left\{ [\nabla \vec{u} + (\nabla \vec{u})^T] \cdot \vec{u}^* \right\}_n^k \\ &= \left\{ 2U_n^k \frac{dU_n^k}{dr} (P_n^k)^2 + V_n^k \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \left[\left(\frac{dP_n^k}{d\theta} \right)^2 + \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \right] \right\} \hat{e}_r \end{aligned}$$

$$\begin{aligned}
& + \left\{ U_n^k \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) P_n^k \frac{dP_n^k}{d\theta} + 2V_n^k \left(\frac{U_n^k}{r} P_n^k + \frac{V_n^k}{r} \frac{d^2 P_n^k}{d\theta^2} \right) \frac{dP_n^k}{d\theta} \right. \\
& \quad \left. + 2 \frac{(V_n^k)^2}{r} \left(-k^2 \frac{\cos \theta}{\sin^3 \theta} P_n^k + \frac{k^2}{\sin^2 \theta} \frac{dP_n^k}{d\theta} \right) P_n^k \right\} \hat{e}_\theta \\
& + \left\{ U_n^k \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \frac{ik}{\sin \theta} (P_n^k)^2 + 2 \frac{(V_n^k)^2}{r} \left(-ik \frac{\cos \theta}{\sin^2 \theta} P_n^k + \frac{ik}{\sin \theta} \frac{dP_n^k}{d\theta} \right) \frac{dP_n^k}{d\theta} \right. \\
& \quad \left. - 2V_n^k \left[\left(\mu \frac{U_n^k}{r} - \frac{V_n^k}{r} \frac{k^2}{\sin^2 \theta} \right) P_n^k + 2 \frac{V_n^k}{r} \cot \theta \frac{dP_n^k}{d\theta} \right] \frac{ik}{\sin \theta} P_n^k \right\} \hat{e}_\lambda \\
& = \left\{ 2U_n^k \frac{dU_n^k}{dr} (P_n^k)^2 + V_n^k \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \left[\left(\frac{dP_n^k}{d\theta} \right)^2 + \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \right] \right\} \hat{e}_r \\
& + \left\{ U_n^k \left(\frac{dV_n^k}{dr} + \frac{U_n^k + V_n^k}{r} \right) P_n^k \frac{dP_n^k}{d\theta} + 2 \frac{(V_n^k)^2}{r} \left[\frac{d^2 P_n^k}{d\theta^2} \frac{dP_n^k}{d\theta} - k^2 \frac{\cos \theta}{\sin^3 \theta} (P_n^k)^2 + \frac{k^2}{\sin^2 \theta} P_n^k \frac{dP_n^k}{d\theta} \right] \right\} \hat{e}_\theta \\
& + \frac{ik}{\sin \theta} \left\{ U_n^k \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) (P_n^k)^2 + 2 \frac{(V_n^k)^2}{r} \left(-\frac{\cos \theta}{\sin \theta} P_n^k + \frac{dP_n^k}{d\theta} \right) \frac{dP_n^k}{d\theta} \right. \\
& \quad \left. - 2V_n^k \left[\left(\frac{U_n^k}{r} - \frac{V_n^k}{r} \frac{k^2}{\sin^2 \theta} \right) P_n^k + 2 \frac{V_n^k}{r} \cot \theta \frac{dP_n^k}{d\theta} \right] P_n^k \right\} \hat{e}_\lambda \tag{22}
\end{aligned}$$

$$\begin{aligned}
\nabla \cdot \{2\vec{\epsilon} \cdot \vec{u}^*\}_n^k &= \frac{2}{r^2} \frac{d}{dr} \left[r^2 U_n^k \frac{dU_n^k}{dr} \right] (P_n^k)^2 + \frac{1}{r^2} \frac{d}{dr} \left[r^2 V_n^k \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \right] \left[\left(\frac{dP_n^k}{d\theta} \right)^2 + \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \right] + \dots \\
& \tag{23}
\end{aligned}$$

where neglected terms do not contribute to result.

$$\begin{aligned}
\{\nabla \cdot (2\vec{\epsilon})\}_n^k &= \left\{ \nabla \cdot [\nabla \vec{u} + (\nabla \vec{u})^T] \right\}_n^k \\
&= \left\{ \frac{d}{dr} \left(2 \frac{dU_n^k}{dr} \right) + \frac{1}{r} \left[4 \frac{dU_n^k}{dr} - 4 \frac{U_n^k}{r} - n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - 3V_n^k}{r} \right) \right] \right\} P_n^k e^{ik\lambda} e^{i\sigma t} \hat{e}_r \\
&+ \left\{ \frac{d}{dr} \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) + \frac{2}{r^2} [U_n^k - (n^2 + n - 1)V_n^k] + \frac{3}{r} \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \right\} \frac{dP_n^k}{d\theta} e^{ik\lambda} e^{i\sigma t} \hat{e}_\theta \\
&+ \left\{ \frac{d}{dr} \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) + \frac{2}{r^2} [U_n^k - (n^2 + n - 1)V_n^k] + \frac{3}{r} \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \right\} \frac{ik}{\sin \theta} P_n^k e^{ik\lambda} e^{i\sigma t} \hat{e}_\lambda \tag{24}
\end{aligned}$$

$$\begin{aligned}
\{[\nabla \cdot (2\vec{\epsilon})] \cdot \vec{u}^*\}_n^k &= \left\{ \nabla \cdot [\nabla \vec{u} + (\nabla \vec{u})^T] \cdot \vec{u}^* \right\}_n^k \\
&= U_n^k \left\{ \frac{d}{dr} \left(2 \frac{dU_n^k}{dr} \right) + \frac{1}{r} \left[4 \frac{dU_n^k}{dr} - 4 \frac{U_n^k}{r} - n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - 3V_n^k}{r} \right) \right] \right\} (P_n^k)^2 \\
&+ V_n^k \left\{ \frac{d}{dr} \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) + \frac{2}{r^2} [U_n^k - (n^2 + n - 1)V_n^k] + \frac{3}{r} \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \right\} \left[\left(\frac{dP_n^k}{d\theta} \right)^2 + \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \right] \tag{25}
\end{aligned}$$

$$\begin{aligned}
& \nabla \cdot (2\vec{\epsilon} \cdot \vec{u}^*) - [\nabla \cdot (2\vec{\epsilon})] \cdot \vec{u}^* \\
&= \left[\frac{2}{r^2} \frac{d}{dr} \left(r^2 U_n^k \frac{dU_n^k}{dr} \right) - U_n^k \frac{d}{dr} \left(2 \frac{dU_n^k}{dr} \right) \right] (P_n^k)^2
\end{aligned}$$

$$\begin{aligned}
& + \left\{ \frac{1}{r^2} \frac{d}{dr} \left[r^2 V_n^k \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \right] - V_n^k \frac{d}{dr} \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \right\} \left[\left(\frac{dP_n^k}{d\theta} \right)^2 + \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \right] \\
& - \frac{U_n^k}{r} \left[4 \frac{dU_n^k}{dr} - 4 \frac{U_n^k}{r} - n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - 3V_n^k}{r} \right) \right] (P_n^k)^2 \\
& - \left\{ \frac{2V_n^k}{r^2} [U_n^k - (n^2 + n - 1)V_n^k] + \frac{3V_n^k}{r} \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \right\} \left[\left(\frac{dP_n^k}{d\theta} \right)^2 + \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \right] \\
& = \left[2 \left(\frac{dU_n^k}{dr} \right)^2 + \frac{4}{r} U_n^k \frac{dU_n^k}{dr} \right] (P_n^k)^2 \\
& + \left\{ \frac{dV_n^k}{dr} \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) + \frac{2V_n^k}{r} \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \right\} \left[\left(\frac{dP_n^k}{d\theta} \right)^2 + \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \right] \\
& - \frac{U_n^k}{r} \left[4 \frac{dU_n^k}{dr} - 4 \frac{U_n^k}{r} - n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - 3V_n^k}{r} \right) \right] (P_n^k)^2 \\
& - \left\{ \frac{2V_n^k}{r^2} [U_n^k - (n^2 + n - 1)V_n^k] + \frac{3V_n^k}{r} \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \right\} \left[\left(\frac{dP_n^k}{d\theta} \right)^2 + \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \right] \\
& = \left\{ 2 \left(\frac{dU_n^k}{dr} \right)^2 + \frac{4}{r} U_n^k \frac{dU_n^k}{dr} - \frac{U_n^k}{r} \left[4 \frac{dU_n^k}{dr} - 4 \frac{U_n^k}{r} - n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - 3V_n^k}{r} \right) \right] \right\} (P_n^k)^2 \\
& + \left\{ \frac{dV_n^k}{dr} \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) - \frac{V_n^k}{r} \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \right\} \left[\left(\frac{dP_n^k}{d\theta} \right)^2 + \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \right] \\
& - \left\{ \frac{2V_n^k}{r^2} [U_n^k - (n^2 + n - 1)V_n^k] \right\} \left[\left(\frac{dP_n^k}{d\theta} \right)^2 + \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \right] \\
& = \left\{ 2 \left(\frac{dU_n^k}{dr} \right)^2 - \frac{U_n^k}{r} \left[-4 \frac{U_n^k}{r} - n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - 3V_n^k}{r} \right) \right] \right\} (P_n^k)^2 \\
& + \left\{ \left(\frac{dV_n^k}{dr} - \frac{V_n^k}{r} \right) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \right\} \left[\left(\frac{dP_n^k}{d\theta} \right)^2 + \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \right] \\
& - \left\{ \frac{2V_n^k}{r^2} [U_n^k - (n^2 + n - 1)V_n^k] \right\} \left[\left(\frac{dP_n^k}{d\theta} \right)^2 + \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \right] \\
& = \left\{ 2 \left(\frac{dU_n^k}{dr} \right)^2 + \frac{U_n^k}{r} \left[\frac{4U_n^k}{r} + n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - 3V_n^k}{r} \right) \right] \right\} (P_n^k)^2 \\
& + \left\{ \left(\frac{dV_n^k}{dr} - \frac{V_n^k}{r} \right) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) - \frac{2V_n^k}{r^2} [U_n^k - (n^2 + n - 1)V_n^k] \right\} \left[\left(\frac{dP_n^k}{d\theta} \right)^2 + \frac{k^2}{\sin^2 \theta} (P_n^k)^2 \right] \quad (26)
\end{aligned}$$

$$\begin{aligned}
K_\mu &= \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta\mu r^2 \left\{ 2 \left(\frac{dU_n^k}{dr} \right)^2 + \frac{U_n^k}{r} \left[\frac{4U_n^k}{r} + n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - 3V_n^k}{r} \right) \right] \right. \\
&\quad \left. + n(n+1) \left(\frac{dV_n^k}{dr} - \frac{V_n^k}{r} \right) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) - n(n+1) \frac{2V_n^k}{r^2} [U_n^k - (n^2 + n - 1)V_n^k] \right\} dr \\
&= \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta\mu r^2 \left\{ 2 \left(\frac{dU_n^k}{dr} \right)^2 + 4 \left(\frac{U_n^k}{r} \right)^2 + n(n+1) \frac{U_n^k}{r} \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) + n(n+1) \frac{U_n^k}{r} \left(\frac{-2V_n^k}{r} \right) \right. \\
&\quad \left. + n(n+1) \left(\frac{dV_n^k}{dr} - \frac{V_n^k}{r} \right) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) - n(n+1) \frac{2V_n^k}{r^2} [U_n^k - (n^2 + n - 1)V_n^k] \right\} dr \\
&= \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta\mu r^2 \left\{ 2 \left(\frac{dU_n^k}{dr} \right)^2 + 4 \left(\frac{U_n^k}{r} \right)^2 + n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right)^2 \right\} dr
\end{aligned}$$

$$\begin{aligned}
& -n(n+1)\frac{2U_n^k V_n^k}{r^2} - n(n+1)\frac{2V_n^k}{r^2} [U_n^k - (n^2 + n - 1)V_n^k] \Big\} dr \\
& = \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta\mu r^2 \left\{ 2 \left(\frac{dU_n^k}{dr} \right)^2 + 4 \left(\frac{U_n^k}{r} \right)^2 + n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right)^2 \right. \\
& \quad \left. - n(n+1)\frac{2V_n^k}{r^2} [2U_n^k - (n^2 + n - 1)V_n^k] \right\} dr
\end{aligned} \tag{27}$$

pair k and μ : $\delta\lambda = \delta k - \frac{2}{3}\delta\mu$

$$\begin{aligned}
K_\mu^k &= \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta\mu r^2 \left\{ 2 \left(\frac{dU_n^k}{dr} \right)^2 + 4 \left(\frac{U_n^k}{r} \right)^2 + n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right)^2 \right. \\
& \quad \left. - n(n+1)\frac{2V_n^k}{r^2} [2U_n^k - (n^2 + n - 1)V_n^k] \right. \\
& \quad \left. - \frac{2}{3} \left[\frac{dU_n^k}{dr} + \frac{2U_n^k}{r} - n(n+1)\frac{V_n^k}{r} \right]^2 \right\} dr \\
&= \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta\mu r^2 \left\{ 2 \left(\frac{dU_n^k}{dr} \right)^2 + 4 \left(\frac{U_n^k}{r} \right)^2 + n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right)^2 \right. \\
& \quad \left. - n(n+1)\frac{2V_n^k}{r^2} [2U_n^k - (n^2 + n - 1)V_n^k] \right. \\
& \quad \left. - \frac{2}{3} \left[\left(\frac{dU_n^k}{dr} \right)^2 + \left(\frac{2U_n^k}{r} - n(n+1)\frac{V_n^k}{r} \right)^2 + 2\frac{dU_n^k}{dr} \left(\frac{2U_n^k}{r} - n(n+1)\frac{V_n^k}{r} \right) \right] \right\} dr \\
&= \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta\mu r^2 \left\{ 2 \left(\frac{dU_n^k}{dr} \right)^2 + 4 \left(\frac{U_n^k}{r} \right)^2 + n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right)^2 \right. \\
& \quad \left. - 4n(n+1)\frac{U_n^k V_n^k}{r^2} + 2 \left[n(n+1)\frac{V_n^k}{r} \right]^2 - 2n(n+1)\frac{(V_n^k)^2}{r^2} \right. \\
& \quad \left. - \frac{1}{3} \left[2 \left(\frac{dU_n^k}{dr} \right)^2 + 2 \left(\frac{2U_n^k}{r} - n(n+1)\frac{V_n^k}{r} \right)^2 + 4\frac{dU_n^k}{dr} \left(\frac{2U_n^k}{r} - n(n+1)\frac{V_n^k}{r} \right) \right] \right\} dr \\
&= \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta\mu r^2 \left\{ 2 \left(\frac{dU_n^k}{dr} \right)^2 + 4 \left(\frac{U_n^k}{r} \right)^2 - 4n(n+1)\frac{U_n^k V_n^k}{r^2} + \left[n(n+1)\frac{V_n^k}{r} \right]^2 \right. \\
& \quad \left. + \left[n(n+1)\frac{V_n^k}{r} \right]^2 - 2n(n+1)\frac{(V_n^k)^2}{r^2} + n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right)^2 \right. \\
& \quad \left. - \frac{1}{3} \left[2 \left(\frac{dU_n^k}{dr} \right)^2 + 2 \left(\frac{2U_n^k}{r} - n(n+1)\frac{V_n^k}{r} \right)^2 + 4\frac{dU_n^k}{dr} \left(\frac{2U_n^k}{r} - n(n+1)\frac{V_n^k}{r} \right) \right] \right\} dr \\
&= \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta\mu r^2 \left\{ 2 \left(\frac{dU_n^k}{dr} \right)^2 + \left[\frac{2U_n^k}{r} - n(n+1)\frac{V_n^k}{r} \right]^2 \right. \\
& \quad \left. + n(n+1) \left(\frac{V_n^k}{r} \right)^2 [n(n+1) - 2] + n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right)^2 \right. \\
& \quad \left. - \frac{1}{3} \left[2 \left(\frac{dU_n^k}{dr} \right)^2 + 2 \left(\frac{2U_n^k}{r} - n(n+1)\frac{V_n^k}{r} \right)^2 + 4\frac{dU_n^k}{dr} \left(\frac{2U_n^k}{r} - n(n+1)\frac{V_n^k}{r} \right) \right] \right\} dr \\
&= \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta\mu r^2 \left\{ \frac{6}{3} \left(\frac{dU_n^k}{dr} \right)^2 + \frac{3}{3} \left[\frac{2U_n^k}{r} - n(n+1)\frac{V_n^k}{r} \right]^2 \right. \\
& \quad \left. + n(n+1) \left(\frac{V_n^k}{r} \right)^2 [n^2 + n - 2] + n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right)^2 \right\} dr
\end{aligned}$$

$$\begin{aligned}
& + \frac{1}{3} \left[-2 \left(\frac{dU_n^k}{dr} \right)^2 - 2 \left(\frac{2U_n^k}{r} - n(n+1) \frac{V_n^k}{r} \right)^2 - 4 \frac{dU_n^k}{dr} \left(\frac{2U_n^k}{r} - n(n+1) \frac{V_n^k}{r} \right) \right] \Bigg\} dr \\
& = \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta\mu r^2 \left\{ (n-1)n(n+1)(n+2) \left(\frac{V_n^k}{r} \right)^2 + n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right)^2 \right. \\
& \quad \left. + \frac{1}{3} \left[4 \left(\frac{dU_n^k}{dr} \right)^2 + \left(\frac{2U_n^k}{r} - n(n+1) \frac{V_n^k}{r} \right)^2 - 4 \frac{dU_n^k}{dr} \left(\frac{2U_n^k}{r} - n(n+1) \frac{V_n^k}{r} \right) \right] \right\} dr \\
& = \frac{4\pi}{2n+1} \frac{(n+k)!}{(n-k)!} \int_0^R \delta\mu r^2 \left\{ \frac{1}{3} \left[2 \frac{dU_n^k}{dr} - \left(\frac{2U_n^k}{r} - n(n+1) \frac{V_n^k}{r} \right) \right]^2 + n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right)^2 \right. \\
& \quad \left. + (n-1)n(n+1)(n+2) \left(\frac{V_n^k}{r} \right)^2 \right\} dr
\end{aligned} \tag{28}$$

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$$\vec{S}_n^k = \left(U_n^k P_n^k \hat{e}_r + V_n^k \frac{dP_n^k}{d\theta} \hat{e}_\theta + V_n^k \frac{ik}{\sin\theta} P_n^k \hat{e}_\lambda \right) e^{ik\lambda}, \tag{29}$$

$$\phi_1 = \Phi_n^k P_n^k e^{ik\lambda} e^{i\sigma t}, \tag{30}$$

$$\bar{T} = \lambda \nabla \cdot \vec{u} \bar{I} + \mu [\nabla \vec{u} + (\nabla \vec{u})^T], \tag{31}$$

$$\nabla \cdot \vec{S}_n^k = \Delta_n^k P_n^k e^{ik\lambda} e^{i\sigma t} \tag{32}$$

$$\Delta_n^k = \frac{dU_n^k}{dr} + \frac{2U_n^k}{r} - n(n+1) \frac{V_n^k}{r}. \tag{33}$$

$$T_{rr} = \left(\lambda \Delta_n^k + 2\mu \frac{dU_n^k}{dr} \right) P_n^k e^{ik\lambda} e^{i\sigma t}, \tag{34}$$

$$T_{\theta\theta} = \left[\left(\lambda \Delta_n^k + 2\mu \frac{U_n^k}{r} \right) P_n^k + 2\mu \frac{V_n^k}{r} \frac{d^2 P_n^k}{d\theta^2} + 2\mu \frac{W_n^k}{r} \left(\frac{ik}{\sin\theta} \frac{dP_n^k}{d\theta} - ik \frac{\cos\theta}{\sin^2\theta} P_n^k \right) \right] e^{ik\lambda} e^{i\sigma t}, \tag{35}$$

$$\begin{aligned}
T_{\lambda\lambda} = & \left[\left(\lambda \Delta_n^k + 2\mu \frac{U_n^k}{r} - 2\mu \frac{V_n^k}{r} \frac{k^2}{\sin^2\theta} \right) P_n^k + 2\mu \frac{V_n^k}{r} \cot\theta \frac{dP_n^k}{d\theta} \right. \\
& \left. - 2\mu \frac{W_n^k}{r} \left(\frac{ik}{\sin\theta} \frac{dP_n^k}{d\theta} - ik \frac{\cos\theta}{\sin^2\theta} P_n^k \right) \right] e^{ik\lambda} e^{i\sigma t},
\end{aligned} \tag{36}$$

$$T_{r\theta} = T_{\theta r} = \left[\mu \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \frac{dP_n^k}{d\theta} + \mu \left(\frac{dW_n^k}{dr} - \frac{W_n^k}{r} \right) \frac{ik}{\sin\theta} P_n^k \right] e^{ik\lambda} e^{i\sigma t}, \tag{37}$$

$$T_{r\lambda} = T_{\lambda r} = \left[\mu \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \frac{ik}{\sin\theta} P_n^k - \mu \left(\frac{dW_n^k}{dr} - \frac{W_n^k}{r} \right) \frac{dP_n^k}{d\theta} \right] e^{ik\lambda} e^{i\sigma t}, \tag{38}$$

$$\begin{aligned}
T_{\theta\lambda} = T_{\lambda\theta} = & \left[2\mu \frac{V_n^k}{r} \left(-ik \frac{\cos\theta}{\sin^2\theta} P_n^k + \frac{ik}{\sin\theta} \frac{dP_n^k}{d\theta} \right) \right. \\
& \left. + \mu \frac{W_n^k}{r} \left(-\frac{d^2 P_n^k}{d\theta^2} + \cot\theta \frac{dP_n^k}{d\theta} - \frac{k^2}{\sin^2\theta} P_n^k \right) \right] e^{ik\lambda} e^{i\sigma t}.
\end{aligned} \tag{39}$$

$$(\nabla \cdot \bar{T})_r = \left\{ \frac{d}{dr} \left(\lambda \Delta_n^k + 2\mu \frac{dU_n^k}{dr} \right) + \frac{\mu}{r} \left[4 \frac{dU_n^k}{dr} - 4 \frac{U_n^k}{r} \right. \right.$$

$$-n(n+1) \left(\frac{dV_n^k}{dr} + \frac{U_n^k - 3V_n^k}{r} \right) \Big] \Big\} P_n^k e^{ik\lambda} e^{i\sigma t}. \quad (40)$$

$$\begin{aligned} (\nabla \cdot \bar{T})_\theta = & \left[\left\{ \frac{d}{dr} \left[\mu \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \right] + \frac{\lambda}{r} \Delta_n^k + \frac{2\mu}{r^2} [U_n^k - (n^2 + n - 1)V_n^k] \right. \right. \\ & + \frac{3\mu}{r} \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \Big\} \frac{dP_n^k}{d\theta} + \left\{ \frac{d}{dr} \left[\mu \left(\frac{dW_n^k}{dr} - \frac{W_n^k}{r} \right) \right] \right. \\ & \left. \left. - (n-1)(n+2) \frac{\mu}{r^2} W_n^k + \frac{3\mu}{r} \left(\frac{dW_n^k}{dr} - \frac{W_n^k}{r} \right) \right\} \frac{ik}{\sin\theta} P_n^k \right] e^{ik\lambda} e^{i\sigma t}. \end{aligned} \quad (41)$$

$$\begin{aligned} (\nabla \cdot \bar{T})_\lambda = & \left[\left\{ \frac{d}{dr} \left[\mu \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \right] + \frac{\lambda}{r} \Delta_n^k + \frac{2\mu}{r^2} [U_n^k - (n^2 + n - 1)V_n^k] \right. \right. \\ & + \frac{3\mu}{r} \left(\frac{dV_n^k}{dr} + \frac{U_n^k - V_n^k}{r} \right) \Big\} \frac{ik}{\sin\theta} P_n^k - \left\{ \frac{d}{dr} \left[\mu \left(\frac{dW_n^k}{dr} - \frac{W_n^k}{r} \right) \right] \right. \\ & \left. \left. - (n-1)(n+2) \frac{\mu}{r^2} W_n^k + \frac{3\mu}{r} \left(\frac{dW_n^k}{dr} - \frac{W_n^k}{r} \right) \right\} \frac{dP_n^k}{d\theta} \right] e^{ik\lambda} e^{i\sigma t}. \end{aligned} \quad (42)$$